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W4 COS30045

Lecture

Class

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Agenda

- 4.1 DOM and SVG
- 4.2 Introduction to D3
- 4.3 Loading data and building charts
- 4.4 Scaling and labelling charts

Agenda

- Drawing SVGs
- Building bar chart with D3

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W4.2 COS30045

Lecture

Agenda

- What is D3?
- D3 basics
- Setting up project

Reference

- Dufour & Meeks (2024) Ch 2

Introduction to D3.js

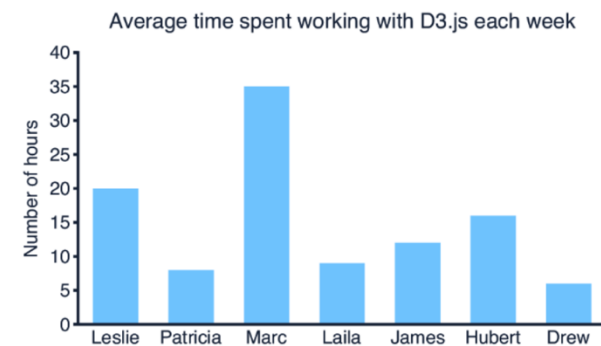
What is D3.js

- Data-Driven Documents
- Open-source JavaScript library
- Web accessible highly customisable interactive storytelling
 - Business
 - Journalism
- Overkill for simple projects
- Not suitable for data processing, analysis or exploration
- Work out your story before starting with D3

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Introduction to D3.js

- Not a charting library
 - Can't tell it to build a 'bar chart'
 - Need to create each element individually



Highcharts

```
Highcharts.chart('container',  
{  
  chart: {  
    type: 'column'  
  },  
  xAxis: {  
    categories: names  
  },  
  yAxis: {  
    title: {  
      text: 'Number of hours'  
    }  
  },  
  series: [{  
    data: [20, 8, 35, 9, 12,  
          16, 6]  
  }]  
});
```

D3.js*

```
d3.csv('../data/my_data.csv')  
  .then(data => {  
    createBarChart(data);  
  });  
  
const createBarChart = (data) => {  
  const xScale = d3.scaleBand()  
    .domain(names)  
    .range([0, width]);  
  const yScale = d3.scaleLinear()  
    .domain([0, 40])  
    .range([height, 0]);  
  
  const svg = d3.select('#viz')  
    .append('svg')  
    .attr('width', width)  
    .attr('height', height);  
  
  const bars = svg  
    .selectAll('rect')  
    .data(data)  
    .join('rect')  
    .attr('x', d => xScale(d.name))  
    .attr('y', d => yScale(+d.num_hours))  
    .attr('width', 60)  
    .attr('height', d => 400 -  
      yScale(+d.num_hours))  
    .attr('fill', '#6a5d7');  
  
  const xAxis = svg  
    .append('g')  
    .call(d3.axisBottom(xScale));  
  const yAxis = svg  
    .append('g')  
    .call(d3.axisLeft(yScale));  
};
```

Load the data.

Create the scales.

Append the SVG parent to the DOM.

Bind the data to rectangles.

Set the position and size of the rectangles.

Append the axes.

* We have simplified this D3.js code for conciseness.

Figure 1.6 A bar chart generated with Highcharts vs. with D3.js. The

Introduction to D3.js

Overview of D3.js project

Note: Current version 7.9.0

- Major changes around version 3 and 4
- Some v3 code won't work with newer versions

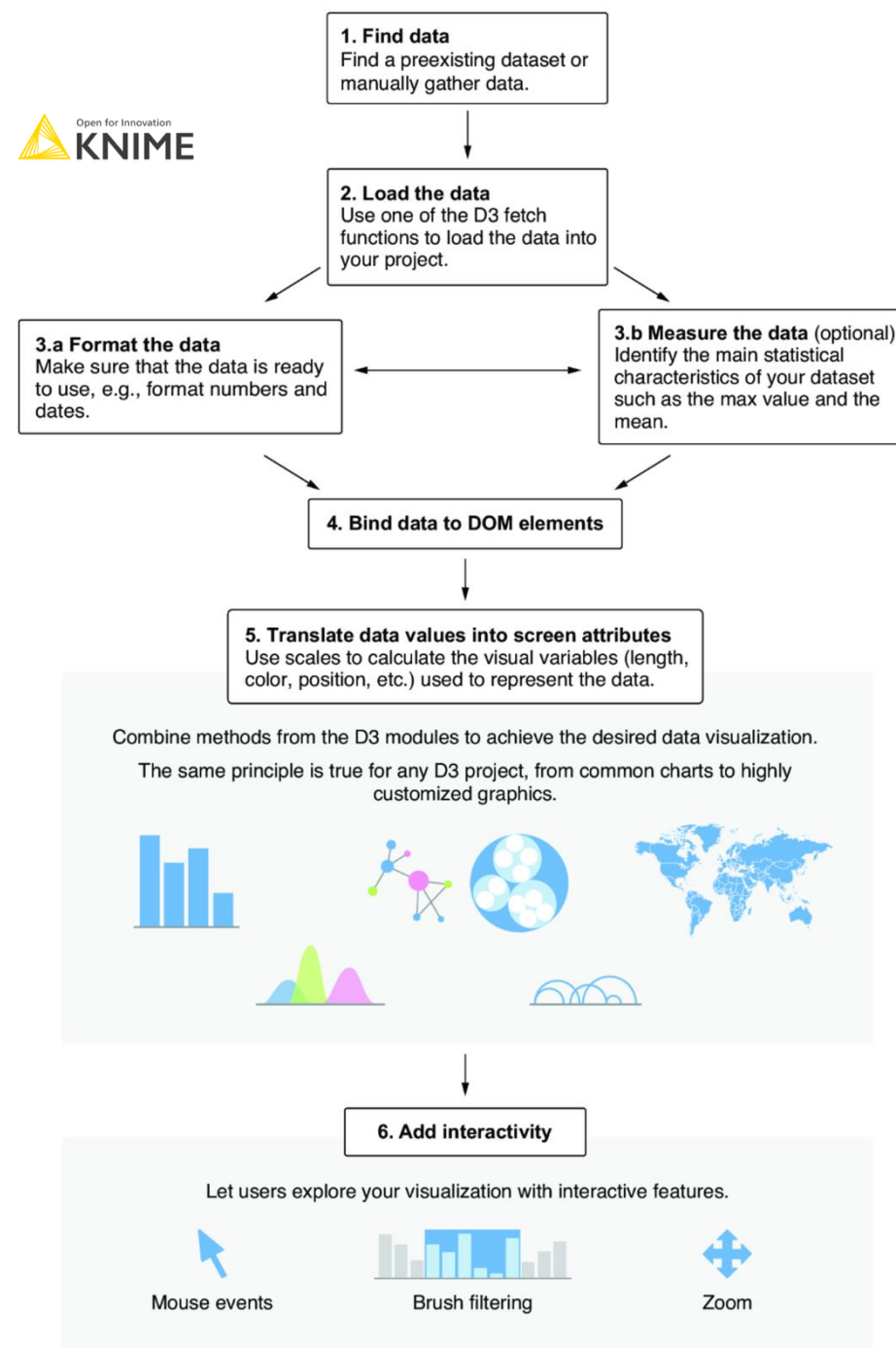
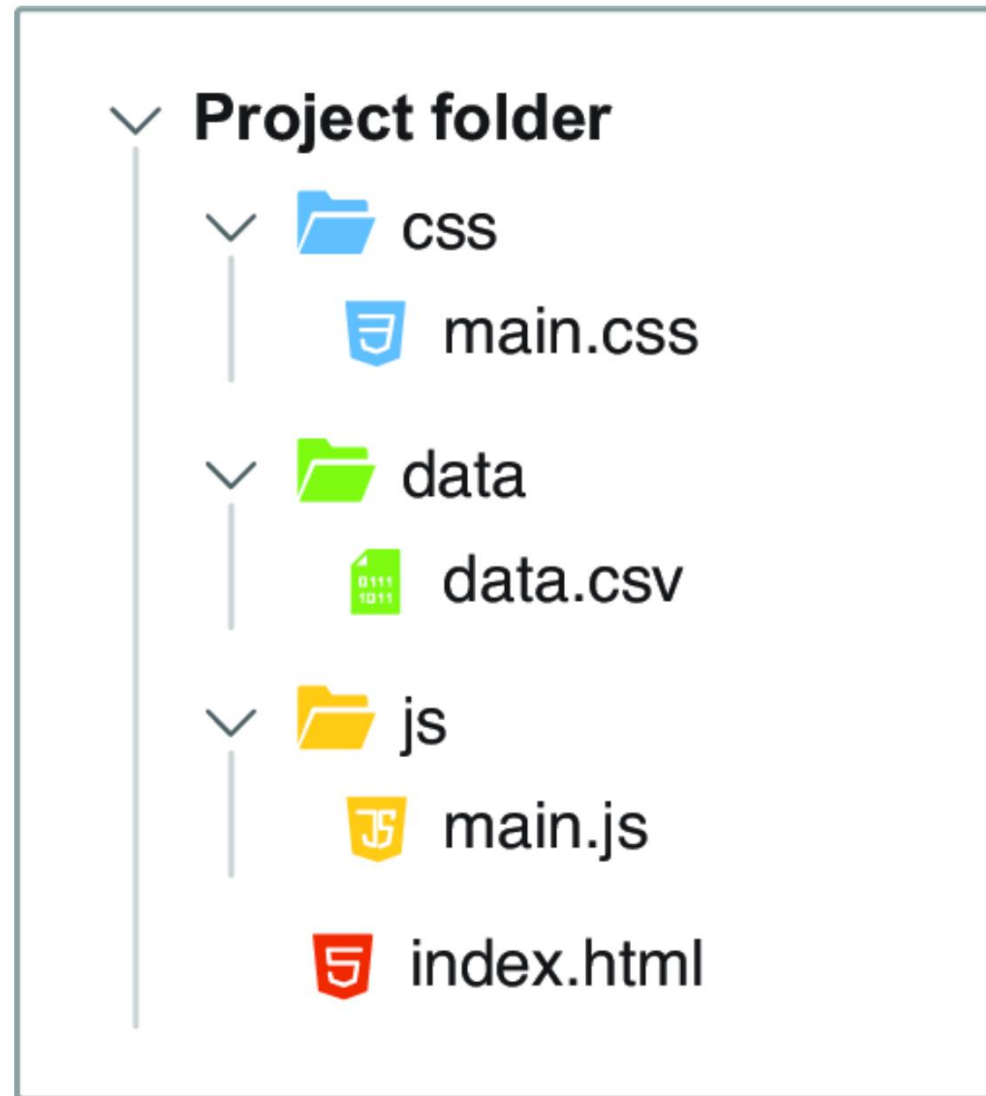


Figure 1.7 How to approach data visualization with D3.js

Building D3 Visualisation



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Figure 2.2 Folder structure of our first D3 project

Index.html – Load D3 Library and your D3.js (i.e., main.js) Code

Call at the end of HTML file to reduce loading time

`</body>`

`<!-- Load your scripts here -->`

`<script src="https://d3js.org/d3.v7.min.js"></script>`

`<script src="js/main.js"></script>`

`</body>`

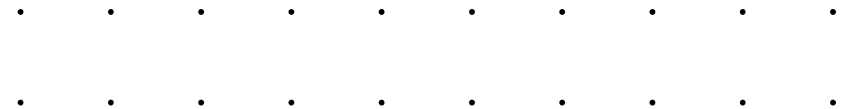
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index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  ...
</head>
<body>
  <div class="container">
    <h1>Wombat populations in Tasmania</h1>
    <p>Wombats can be found in many Tasmanian habitats.</p>
  </div>

  <script src="https://d3js.org/d3.v7.min.js"></script>
  <script src="js/main.js"></script>

</body>
</html>
```



Selecting elements

`d3.select();`

`d3.selectAll();`

`d3.select("selector");`

For example:

`d3.select("p");`

`d3.select("svg");`

`d3.select("#my-id");`

`d3.selectAll("p");`

```
d3.select("selector");
```

	(string)
ID	"#my-id"
class	".my-class"
tag name	"h1" "svg"
combination of selectors	"circle.faded" "#my-id p" "rect, circle"

Figure 2.4 The `select()` method

Selecting elements in DOM

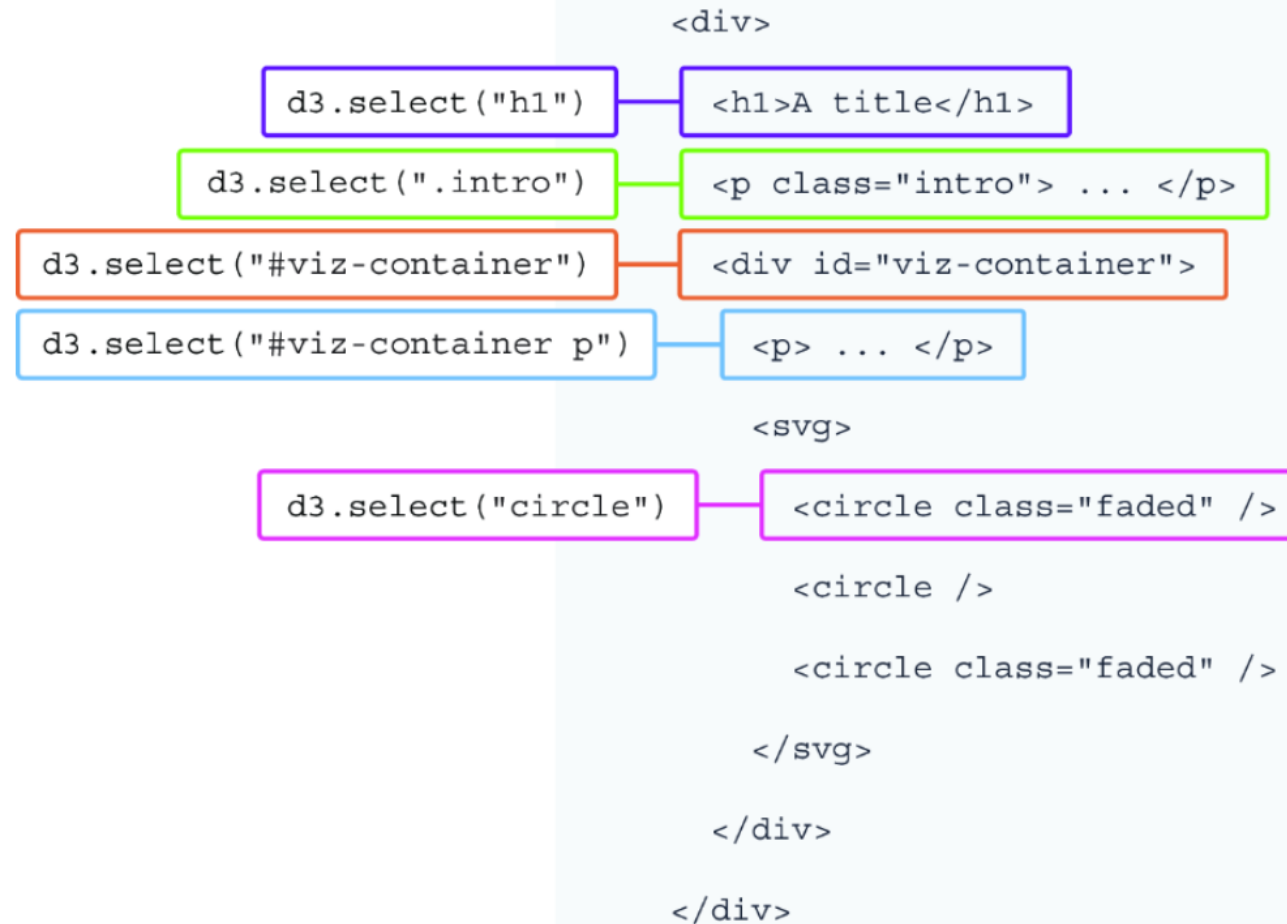


Figure 2.6 DOM elements returned by the `d3.select()` method

```
webpage-project > src > index.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Webpage Project</title>
7   <link rel="stylesheet" href="css/styles.css">
8 </head>
9 <body>
10   <div class="container">
11     <h1>Wombat populations in Tasmania</h1>
12     <p>Wombats can be found in many Tasmanian habitats.</p>
13   </div>
14   <script src="https://d3js.org/d3.v7.min.js"></script>
15   <script src="js/main.js"></script>
16
17 </body>
18 </html>
```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

```
webpage-project > src > css > # styles.css > ...
1 body {
2   font-family: Arial, sans-serif;
3   margin: 0;
4   padding: 20px;
5   /* background-color: #f4f4f4; */
6 }
7
8 h1 {
9   color: #333;
10  text-align: center;
11 }
12
13 p {
14   color: #666;
15   font-size: 16px;
16   line-height: 1.5;
17 }
```

styles.css

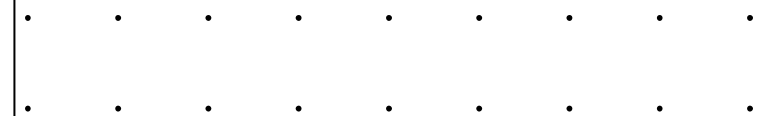
main.js - add some D3 code

```
d3.select("h1").style("color", "green");
```

```
webpage-project > src > js > JS main.js
1
2
3 d3.select("h1").style("color", "green");
4
5
```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.



Selecting elements in DOM

`d3.selectAll()`

To select all circles



Figure 2.7 DOM elements returned by the `d3.selectAll()` method

Html with no D3

```
WombatTest > <> index.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  > <head>...
4  </head>
5  <body>
6  <div>
7  <h1>Wombat populations in Tasmania</h1>
8  <div class="container">
9  <p class="intro">Wombats can be found in many Tasmanian habitats.</p>
10 <div id="viz-container">
11 <p>Here is a chart showing their distribution.</p>
12 <svg>
13 <circle class="faded" cx="50" cy="50" r="40" fill="red"></circle>
14 <circle cx="110" cy="45" r="15" fill="green"></circle>
15 <circle class="faded" cx="100" cy="100" r="30" fill="brown"></circle>
16 </svg>
17 </div>
18 </div>
19 </body>
20 </html>
```

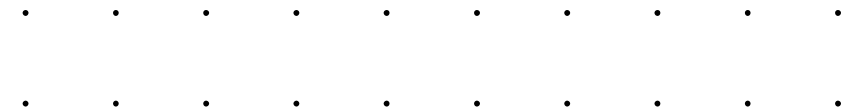
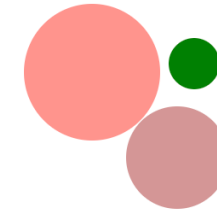
style.css

```
.faded {
  opacity: 0.5; /* Makes the element 50% transparent */
  transition: opacity 0.3s ease; /* Adds a smooth transition effect */
}
```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

Here is a chart showing their distribution.





main.js – using D3 to manipulate DOM

```

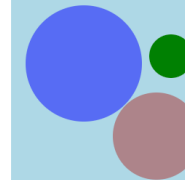
1
2 d3.select("h1").style("color", "green");
3
4 d3.select(".intro")
5   .style("color", "darkblue");
6
7 d3.select("#viz-container")
8   .style("background-color", "lightblue");
9
10 d3.select("circle")
11   .style("fill", "blue");
12
13

```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

Here is a chart showing their distribution.



```

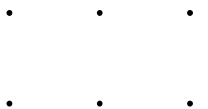
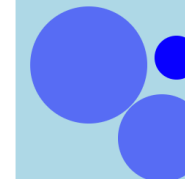
2 d3.select("h1").style("color", "green");
3
4 d3.select(".intro")
5   .style("color", "darkblue");
6
7 d3.select("#viz-container")
8   .style("background-color", "lightblue");
9
10 d3.selectAll("circle")
11   .style("fill", "blue");
12
13

```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

Here is a chart showing their distribution.



Adding elements to a selection

Use D3 to add elements to selection using **append()**.

For example,

```
d3.select("svg")  
    .append("rect")
```

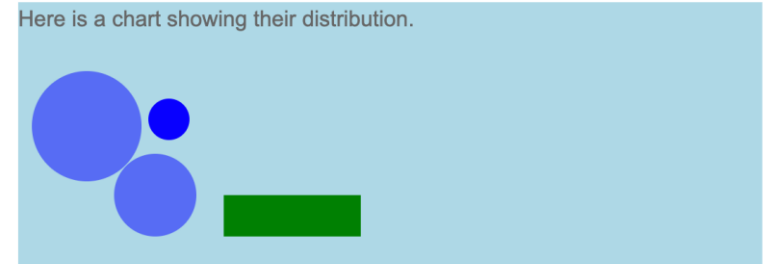
New DOM element created

```
1  
2 d3.select("h1").style("color", "green");  
3  
4 d3.select(".intro")  
5   .style("color", "darkblue");  
6  
7 d3.select("#viz-container")  
8   .style("background-color", "lightblue");  
9  
10 d3.selectAll("circle")  
11   .style("fill", "blue");  
12  
13 d3.select("svg")  
14   .append("rect")  
15     .attr("x", 150)  
16     .attr("y", 100)  
17     .attr("width", 100)  
18     .attr("height", 30)  
19     .style("fill", "green");  
20
```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

Here is a chart showing their distribution.



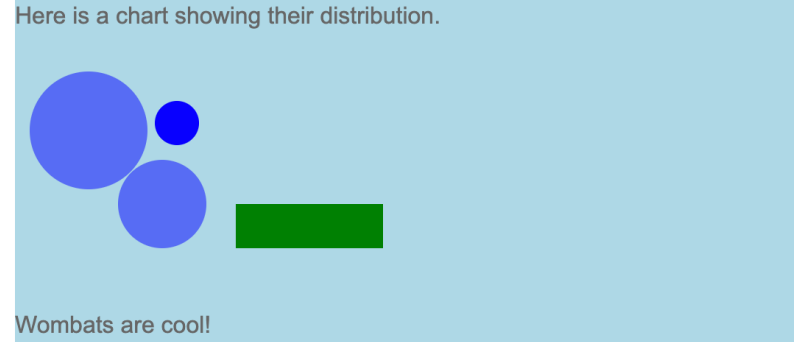
Append

```
1
2 d3.select("h1").style("color", "green");
3
4 d3.select(".intro")
5   .style("color", "darkblue");
6
7 d3.select("#viz-container")
8   .style("background-color", "lightblue");
9
10 d3.selectAll("circle")
11   .style("fill", "blue");
12
13 d3.select("svg")
14   .append("rect")
15     .attr("x", 150)
16     .attr("y", 100)
17     .attr("width", 100)
18     .attr("height", 30)
19     .style("fill", "green");
20
21 d3.selectAll("div")
22   .append("p")
23     .text("Wombats are cool!");
```

Wombat populations in Tasmania

Wombats can be found in many Tasmanian habitats.

Here is a chart showing their distribution.



Wombats are cool!

Wombats are cool!

Wombats are cool!

Three divs,
three cool wombats!

```
2 <html lang="en">
9 <body>
10   <div>
13     <div class="container">
17       <div id="viz-container">
18         <p>Here is a chart showing their distribution.</p>
19
20         <svg>
21           <circle class="faded" cx="50" cy="50" r="40" fill="red"></circle>
22           <circle cx="110" cy="45" r="15" fill="green"></circle>
23           <circle class="faded" cx="100" cy="100" r="30" fill="brown"></circle>
24         </svg>
25       </div>
26     </div>
27   </div>
28
29 </div>
30 <script src="https://d3js.org/d3.v7.min.js"></script>
31 <script src="js/main.js"></script>
32
33 </body>
34 </html>
```

Setting up D3 project

- Load D3 library
- Load main.js script
- Create a responsive container div
- Select and append svg
- Give svg a viewBox attribute
- Add rectangle (first bar in our chart)

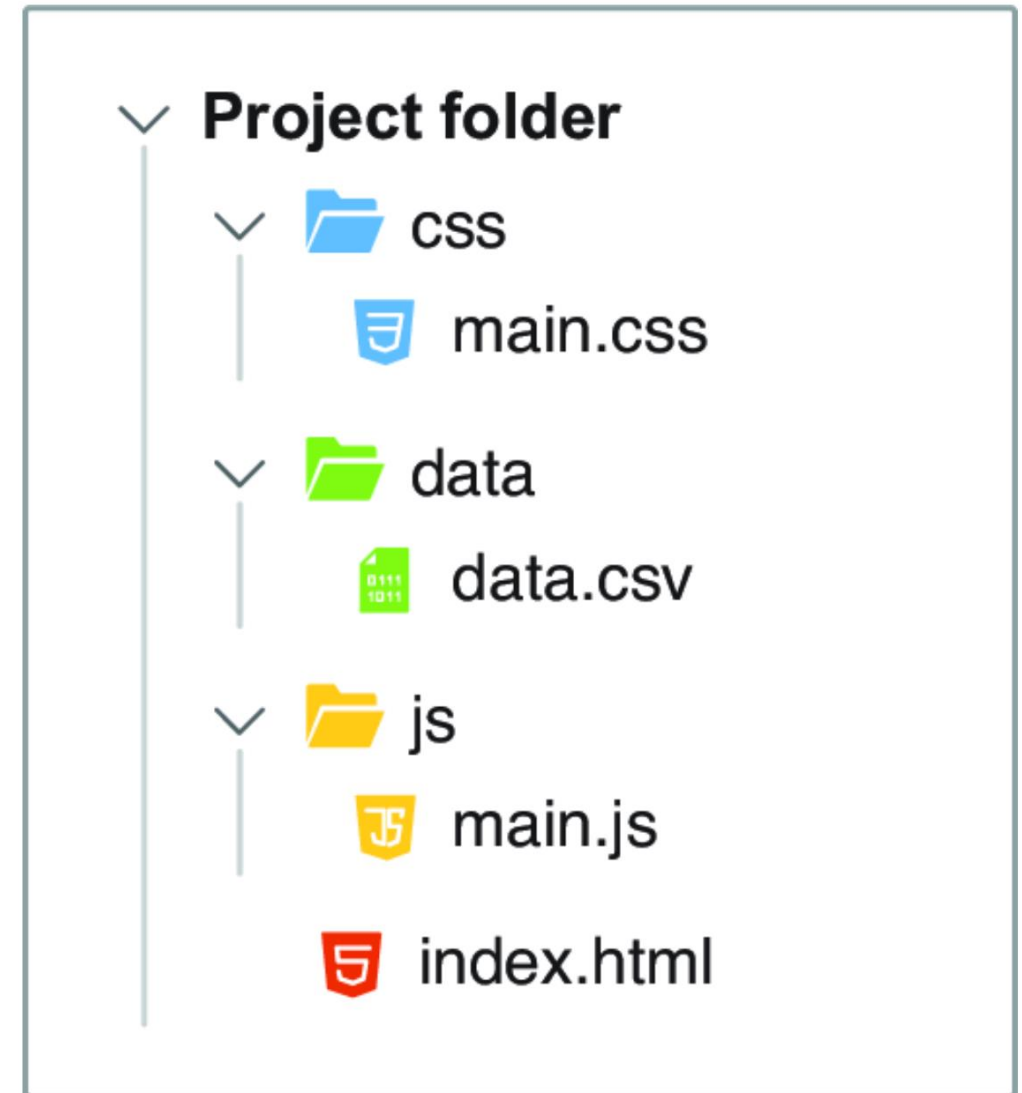


Figure 2.2 Folder structure of our first D3 project

Index.html - responsive-svg-container

Makes sure svg scales nicely

```
<body>
```

```
<div class="responsive-svg-container"></div>
```

```
<script src="https://d3js.org/d3.v7.min.js"></script>
```

```
<script src="js/main.js"></script>
```

```
</body>
```

CSS file

```
.responsive-svg-container {  
  margin-right: auto;  
  margin-left: auto;  
  width: 100%;  
  max-width: 1200px;  
}
```

main.js - Add svg canvas to page inside responsive-svg-container div

```
const svg = d3.select(".responsive-svg-container")
```

```
.append("svg")
```

origin width and height

```
.attr("viewBox", "0 0 1200 1600")
```

```
.style("border", "1px solid black");
```

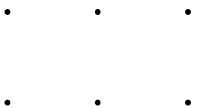
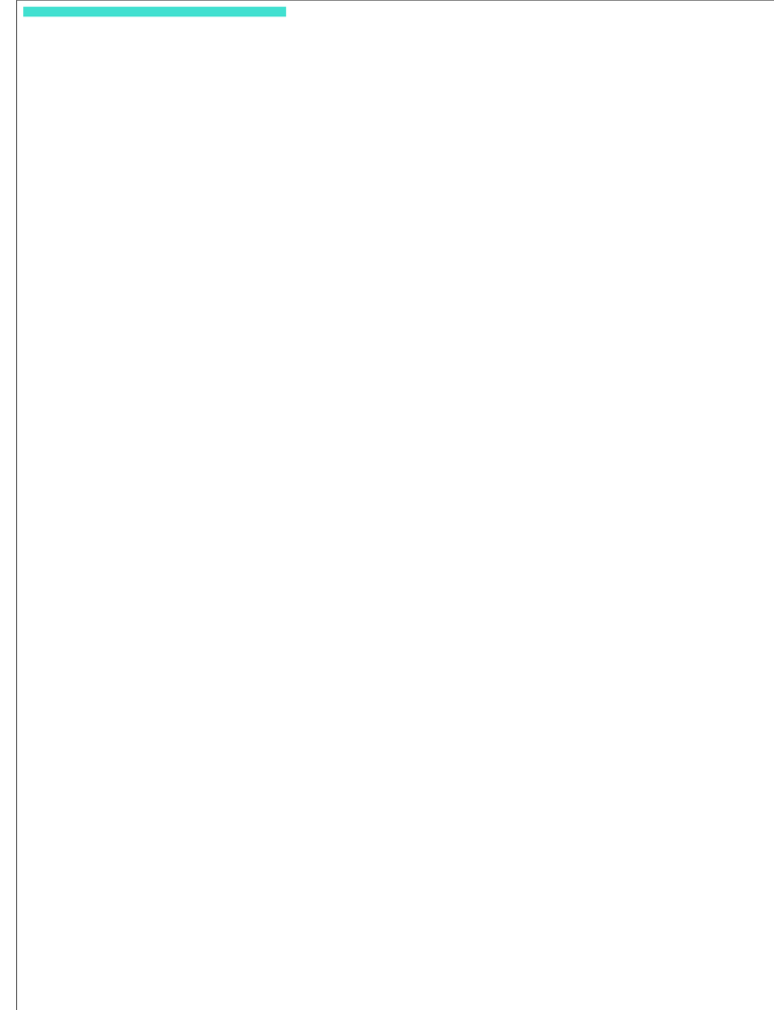
Note: camel case
on attribute

Visualisation will display relative to svg
SVGs elements with co-ordinates off the svg will not be displayed

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Add rectangle

```
svg
  .append("rect")
    .attr("x", 10)
    .attr("y", 10)
    .attr("width", 414)
    .attr("height", 16)
    .attr("fill", "turquoise");
```



Setting and selecting attributes

Use D3 to modify selections using `attr()`. For example,

```
.append("rect")  
  .attr("x", 10)  
  .attr("y", 10)  
  .attr("width", 400)  
  .attr("height", 20)  
  .attr("fill", "turquoise");
```

```
.append("rect")  
  .attr("x", 25)  
  .attr("y", 50)  
  .attr("height", 50)  
  .attr("width", 200)  
  .attr("fill", green);
```

```
. . . . .  
. . . . .
```

Selecting and modifying styles

`.style` will overwrite css and attribute

```
rect {  
    fill: orange;  
}
```

```
.style("fill", "plum");
```

```
.attr("fill", "turquoise")
```

Not applied in order

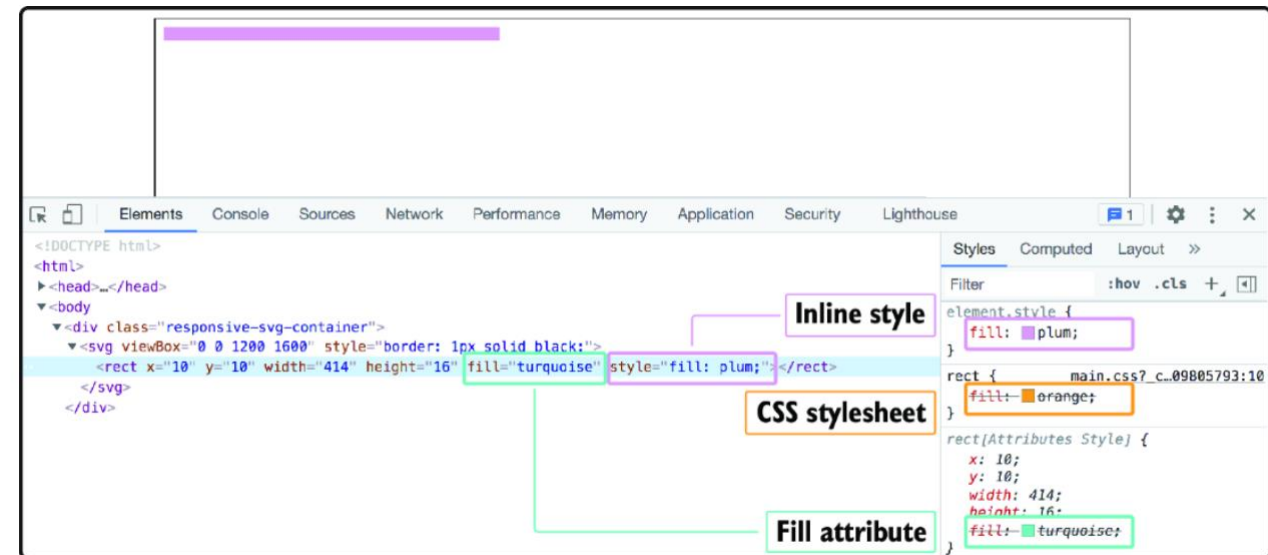


Figure 2.18 Fill applied as an attribute, from an external stylesheet, and as an inline style

D3 data workflow

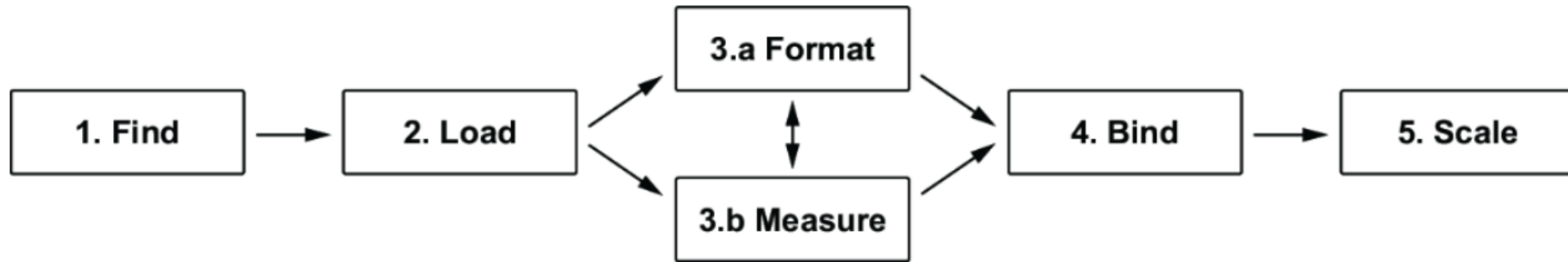


Figure 3.2 Simplified diagram of the D3 data workflow

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